CS & IT ENGINEERING



Algorithm

Miscellaneous

Lecture No. 02



By- Aditya sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

TC using Recurrence Tree

Topics to be Covered



- Topic
- Topic
- Topic

Set
- Union
- Intersection

Kruskal → (Heap + Set)

About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.





Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW



Topic: Sets

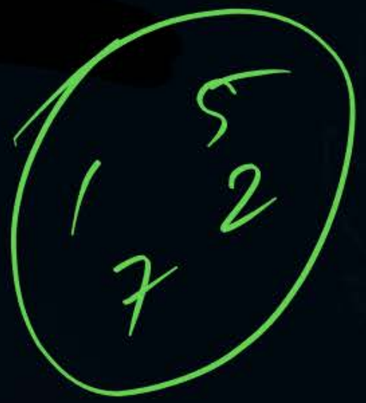
Unordered

~~Wonder~~ collection of non- repeated/ non- duplicate/ distinct elements.

Example:

$S_1 = \{1, 5, 7, 2\}$







Topic: Sets

~~Veen~~ diagram:

1. Disjoint sets: Mutually exclusive sets

   number common elements = 0

Equation:

S1: set of even numbers = {2, 4, 6}

S2: set of ~~ever~~ odd numbers = {1, 3, 5}

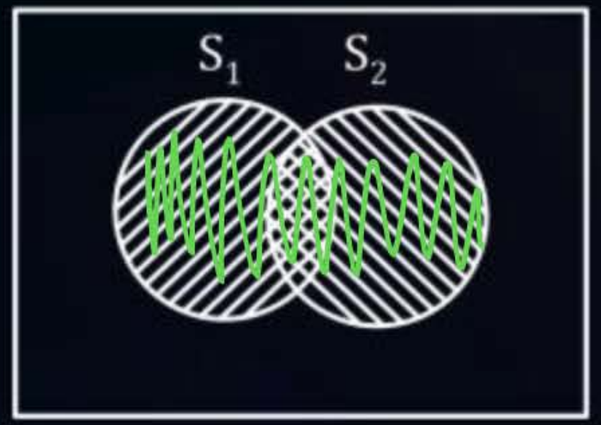
$S1 \cap S2 = \phi$ (empty set)



Topic: Sets

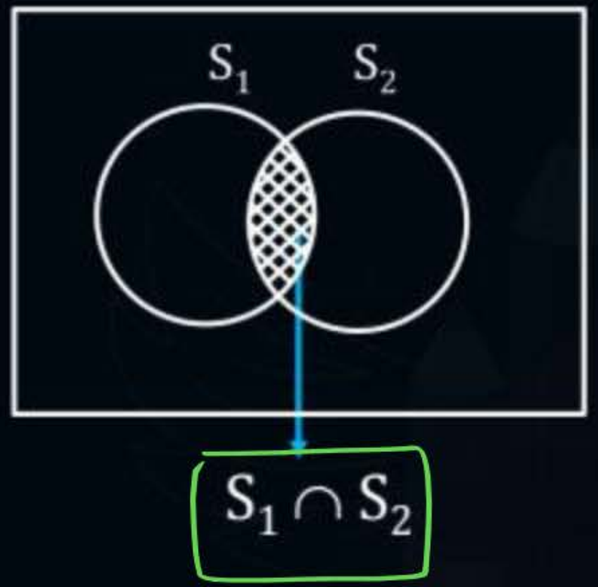
1. Union:

$S_1 \cup S_2 = \{1, 2, 3, 4, 5, 6\}$



2. Intersection

$S_1 \cap S_2$

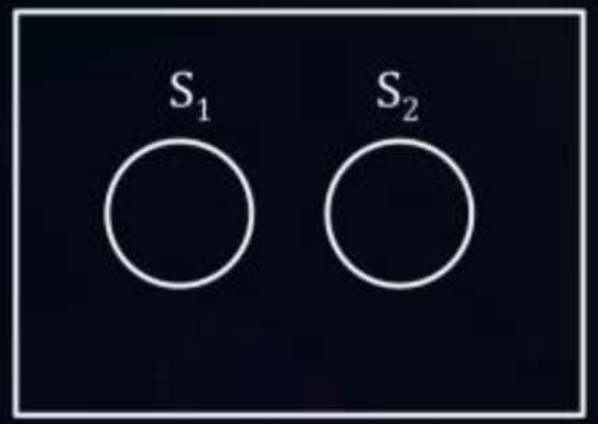




Topic: Sets

3. Disjoint sets

$S_1 \cup S_2 = \{1, 2, 3, 4, 5, 6\}$



$S_1 \cap S_2 = \phi = \{\}$



Topic: Sets

3. Set Difference: $S_1 - S_2$

Equation:

S_1: {2, 3, 4}

S_2: {3, 5, 7}

Element is S1 that are not is S2.

$S_1 - S_2 = \{2,3,4\} - \{3\}$

$= \{2, 4\}$

$S_1 - S_2 = S_2 - S_1$?

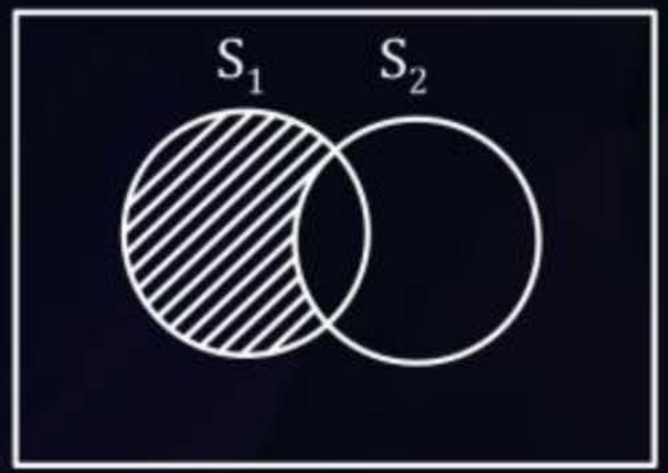


Topic: Sets

Is $S_1 - S_2 = S_2 - S_1$

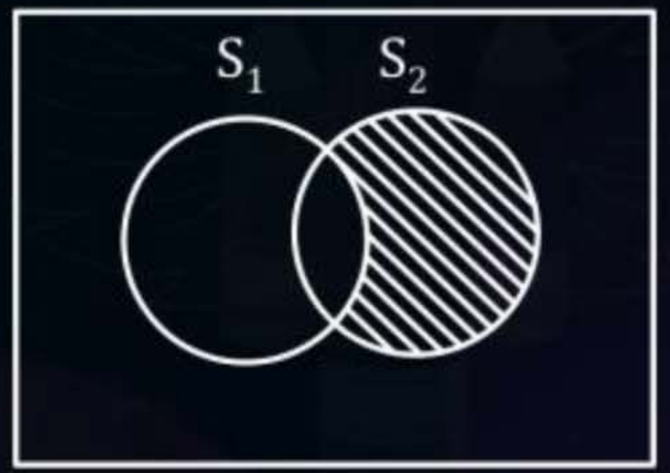
Not necessary

$S_1 - S_2$



$\neq$

$S_2 - S_1$





Topic: Sets

1) It is assumed that the elements of the sets are the numbers 1, 2, 3, n. These number might, in practice, be indices into a symbol table in which the names of the elements are stored. We assume that the sets being represented are pairwise disjoint (that is, if S_i, and s_j, $i \neq j$, are two sets, then there is no element that is in both S_i and S_j).

2) For example, when n = 10 the elements can be partitioned into three disjoint sets, $S_1 = \{1, 7, 8, 9\}$, $S_2 = \{2, 5, 10\}$, and $S_2 = \{3,4, 6\}$. representation for these





Topic: Sets

Tree Diagram Representation of a set:

S_1 : {1, 7, 8, 9}

S_2 : {2, 5, 10 }

S_3 : {3, 4, 6}



Find (8) =1

Find (1) =1

Find (3) =3

Find (7) =1

Find (10) =5



Topic: Sets

Important points:

1. In tree diagram representation of a set, the Root of the tree is considered as the representative of the set.
2. Usually (not mandatory), that the 1st element of the set is the representative of that set.



Topic: Sets

V. Imp

The operations we wish to perform on these sets are:

1. Disjoint set union. If S_i and S_j are two disjoint sets, then their union S_i U S_j = all element x such that x is in S_i or S_j. Thus $S_1 \cup S_2$ = {1, 7, 8, 9, 2, 5, 10). Since we have assumed that all sets are disjoint, we can assume that following the union of S_i and S_j the sets S_i and S_j do not exist independently; that is, they are replaced by $S_i \cup S_j$ in the collection of sets.
2. Find (i). Given the element i, find the set containing i. Thus, 4 is in set S_3, and 9 in set S_1.

Representative

$S_1 = \{1\}$

$S_2 = \{2\}$

S_3

$S_1 \cup S_2 = \{1, 2\}$



Topic: Sets

Party-1 → R1

Party-2 → R2

Party-1 + Party-2

R1 or R2

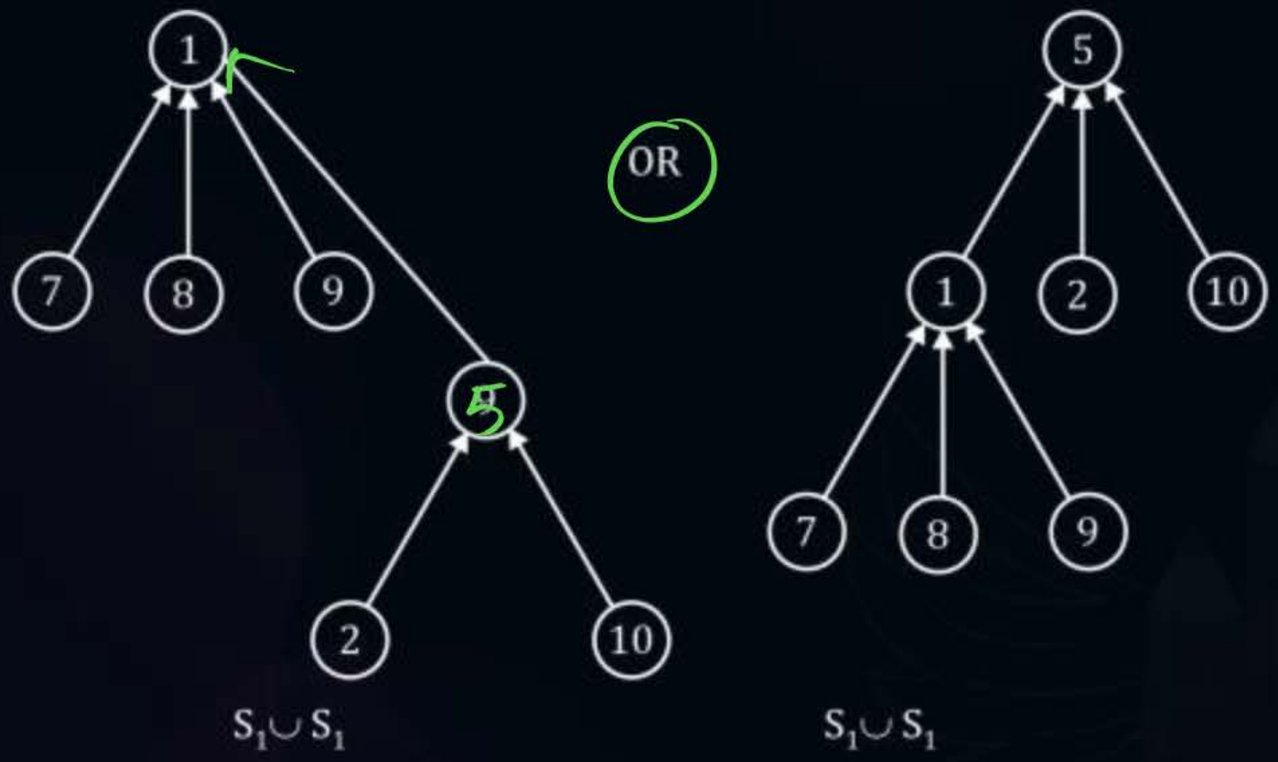


Topic: Sets

V. imp

S_1: {1, 7, 8, 9}

S_1: {5, 2, 10}



$S_1 \cup S_1$ $S_1 \cup S_1$

Possible representation of $S_1 \cup S_2$



Topic: Sets

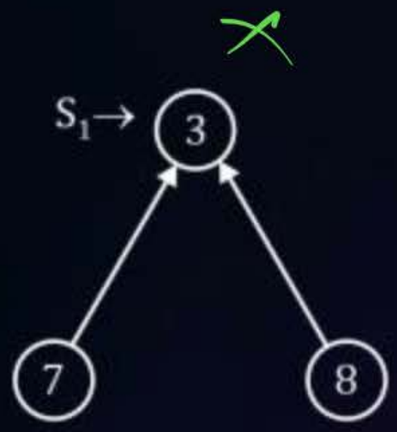
V. Imp

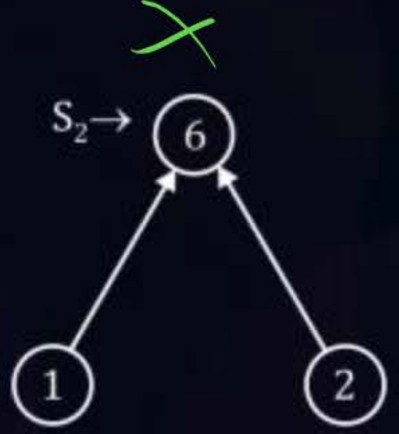
Important:

Difference between parent and find/ Representative

S_1: {3, 7, 8}

S_1: {6, 1, 2}





$S_1 \cup S_2$



Representative

Find (8) = 3

Find (6) = 3

Find (2) = 3

Find (i) always gives the root

Representation



Topic: Sets

Important Role:

In above $S_1 \cup S_2$

6 is the parent of 2 but 3 is the overall set representative.

Parent (2) = 6

Find (2) = 3

Parent (Root) = –1

Parent (3) = –1

Find (3) : 3



Topic: Sets

Parent (Representative) = –1

Parent (Root) = –1

i	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]
P	–1	5	–1	3	–1	3	1	1	1	5

Array representation of S_1, S_2 and S_3 of figure

S_1: {1, 7, 8, 9}

S_2: {2, 5, 10}

S_3: {3, 4, 6}



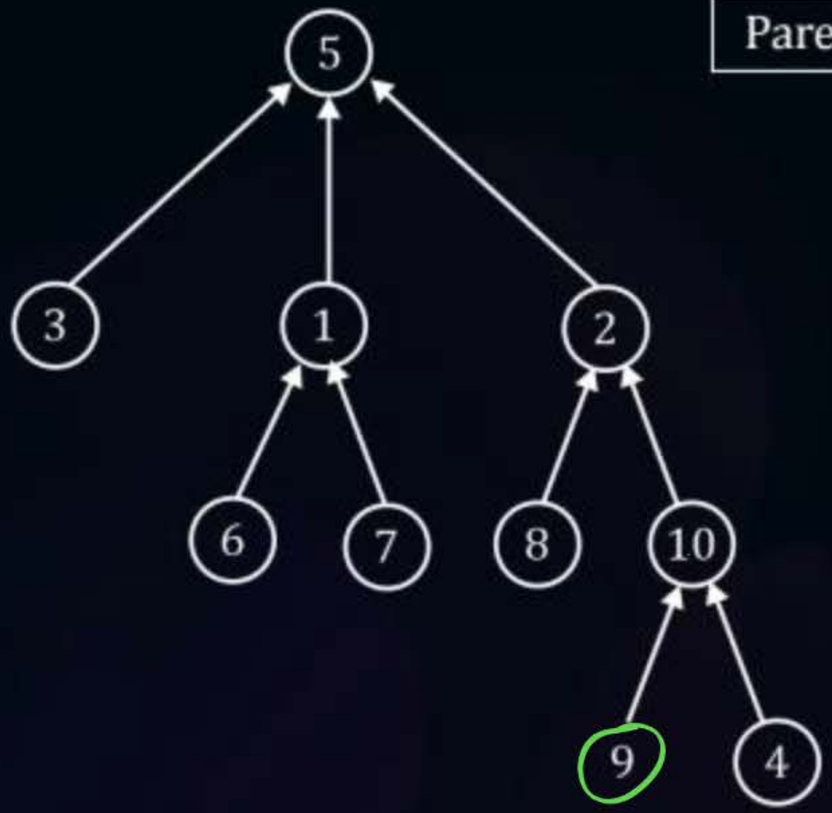
[NAT]

#Q. Write down the array representation of the given tree diagram of set.



Topic: Sets

Solution:



i	1	2	3	4	5	6	7	8	9	10
Parent [i]	5	5	5	10	-1	1	1	2	10	2

Find (4) = 5
Parent (4) =10

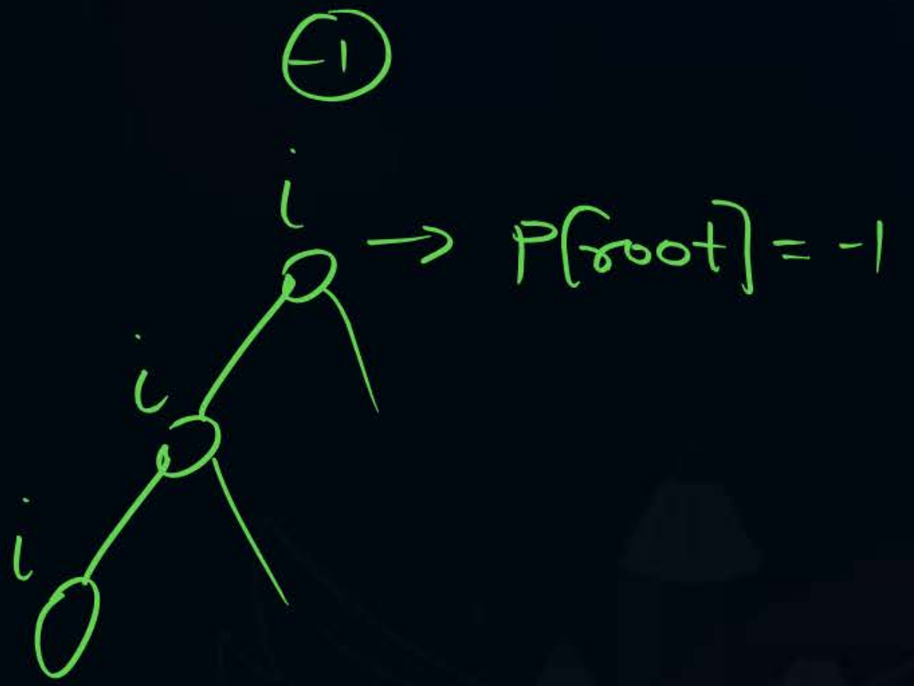
Find (5) = 5
Parent ~~Find~~ (5) = -1



Topic: Sets

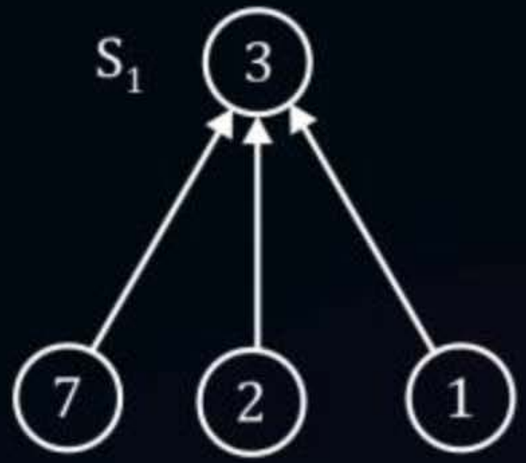
```
1. Algorithm Union (i, j)
2. {
3.       P[i] = j;
4. }
```

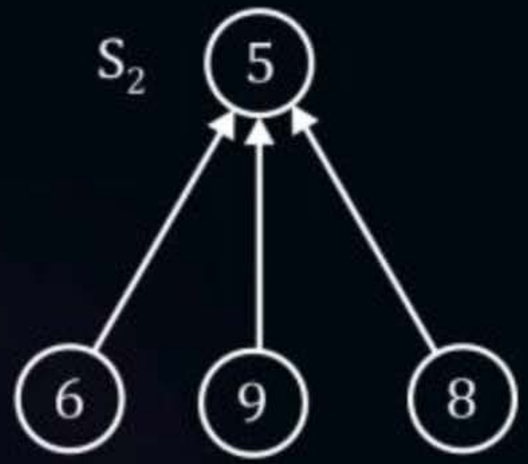
```
1. Algorithm find (i)
2. {
3.       While (p [i]≥ 0) do i=p [i];
4.       return i
5. }
```

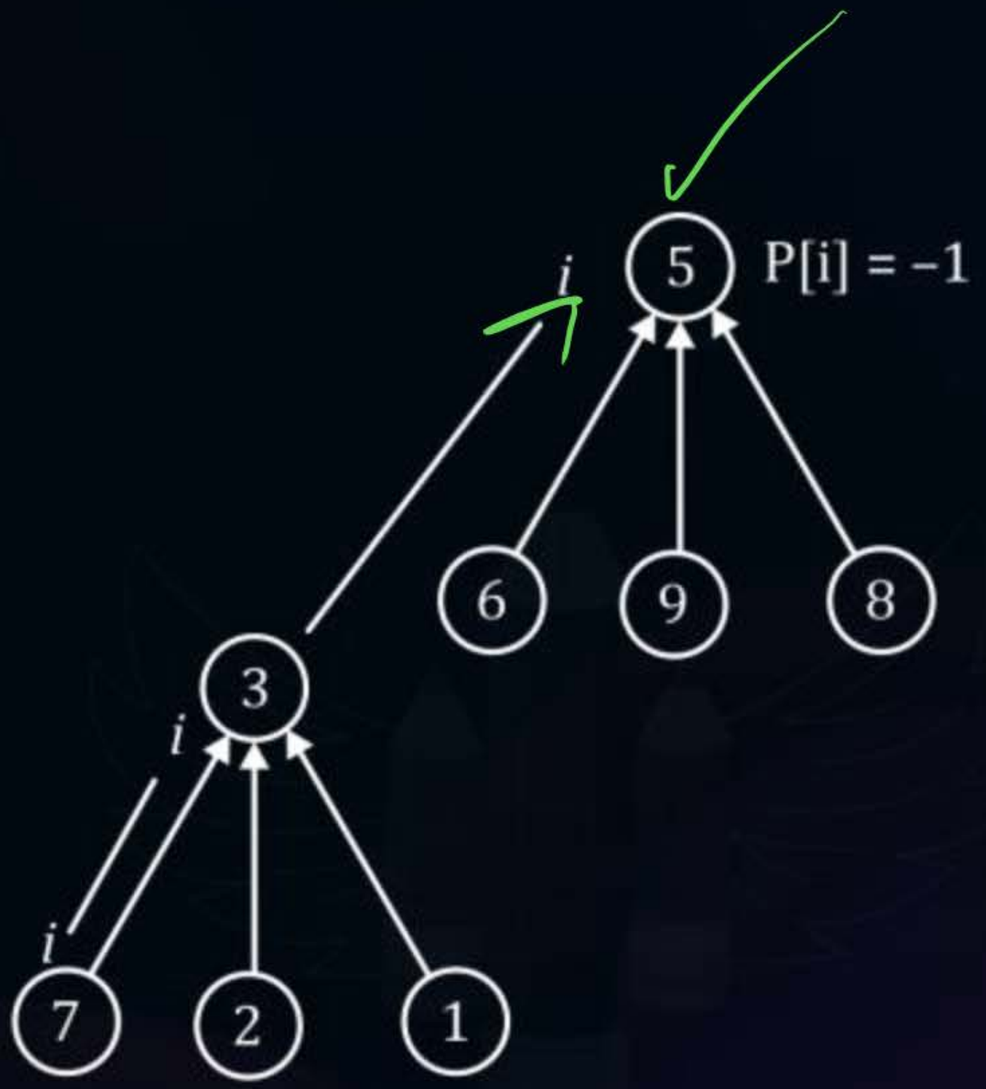




Topic: Sets









Topic: Sets

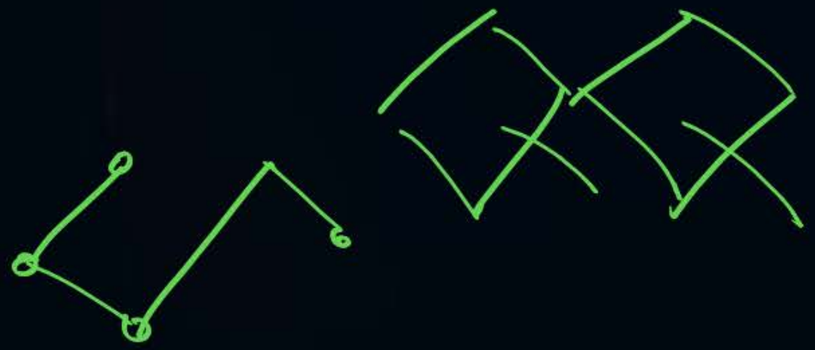
i = 7

Find (7)

- i = 7 P[i]= 3 ≥ 0
- i = 3 P[3]= 5 ≥ 0
- i = 5 P[5]= –1 ≥ 0 (False)
- exit loop
- return (5)



Topic: Greedy Method



```
1.  Algorithm Kruskal [E, cost, n, t]
2.  {
3.  Construct a min Heap out of the edge cost using Heapify
4.  For i = 1 to n do parent [i] : = –1;
5.  i : 0; mincost: = 0.0;
6.  While ((i< n –1) and (heap not empty)) do
7.  {
8.  Delete a minimum cost edge (u,v) from the heap and reheapify using adjust;
9.  J = find (u);k = find (v);
10. If (j≠k then)
11. {
12. i: i + 1
```

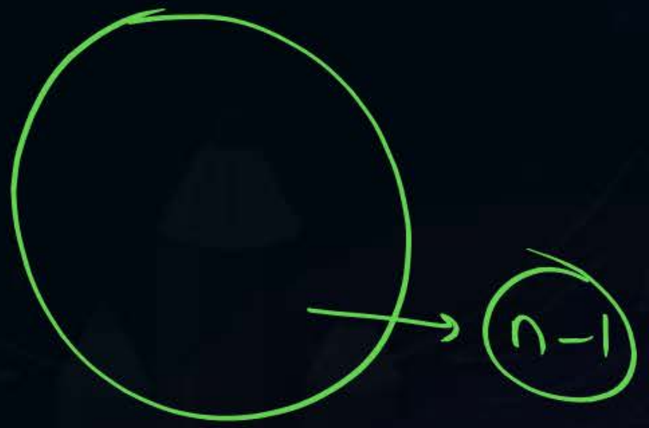




Topic: Greedy Method

```
13.     T[i,1]:=u; [i, 2] =v
14.     Mincost := mincost + cost [u, v]
15.         Uninon (j ,k)
16.       }
17.   }
18.       If (i ≠ n–1) then write (“No spanning tree”)
19.       Else return mincost;
20.   }
```

t[i,1] = u, t[i,2] = v



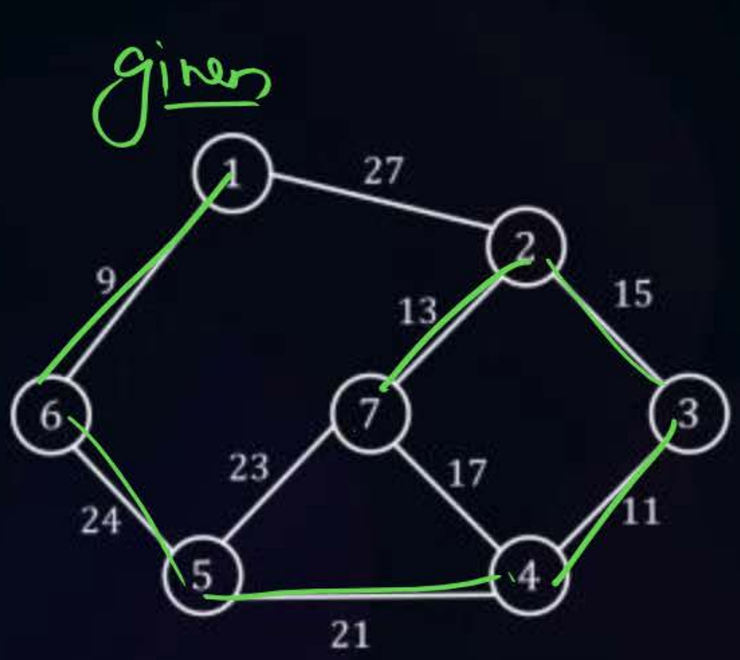


Topic: Sets

Code walk through:

Given graph: edge lost: [9, 11, 13, 15, 17, 21, 23, 24, 27]

Min heap of edge cots of given graph.



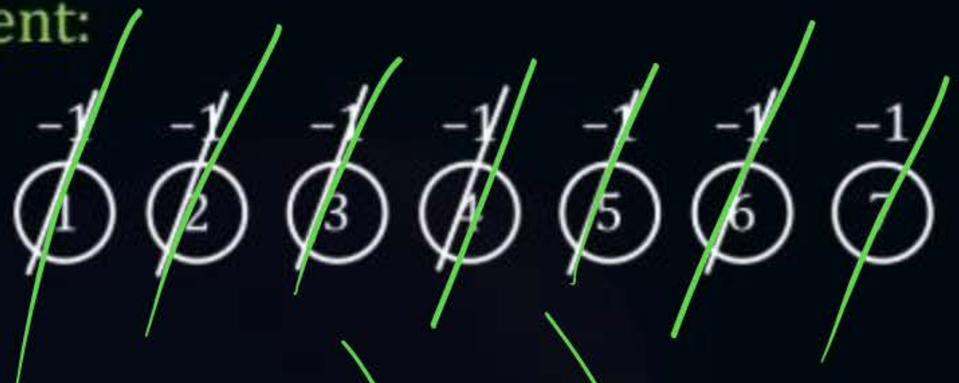


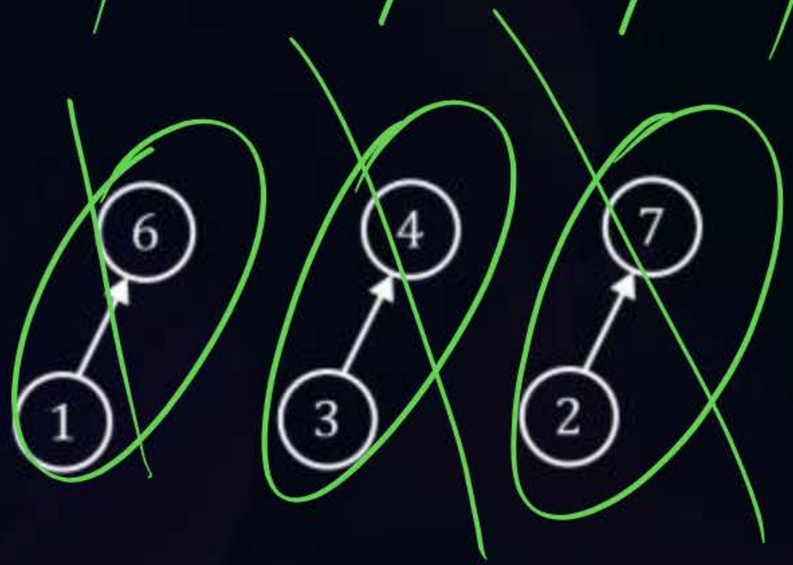


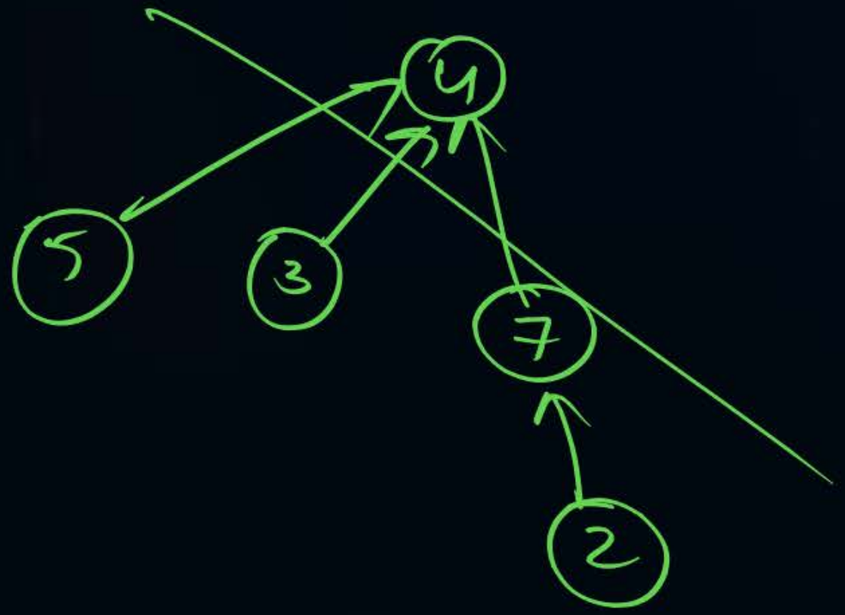
Topic: Sets

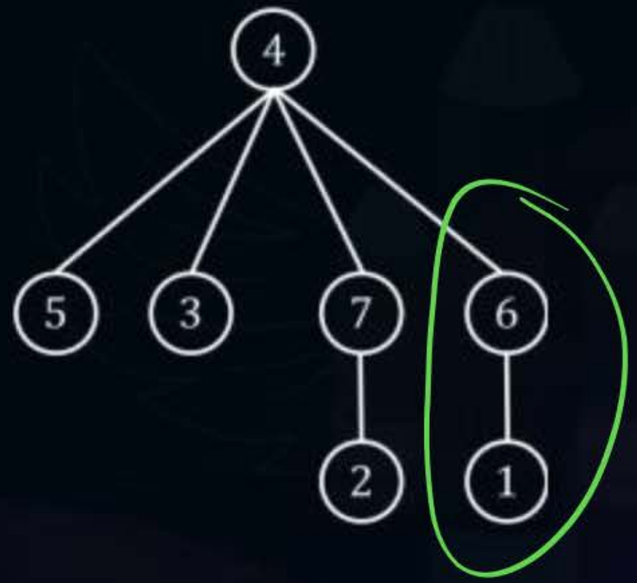
Initially

Parent:





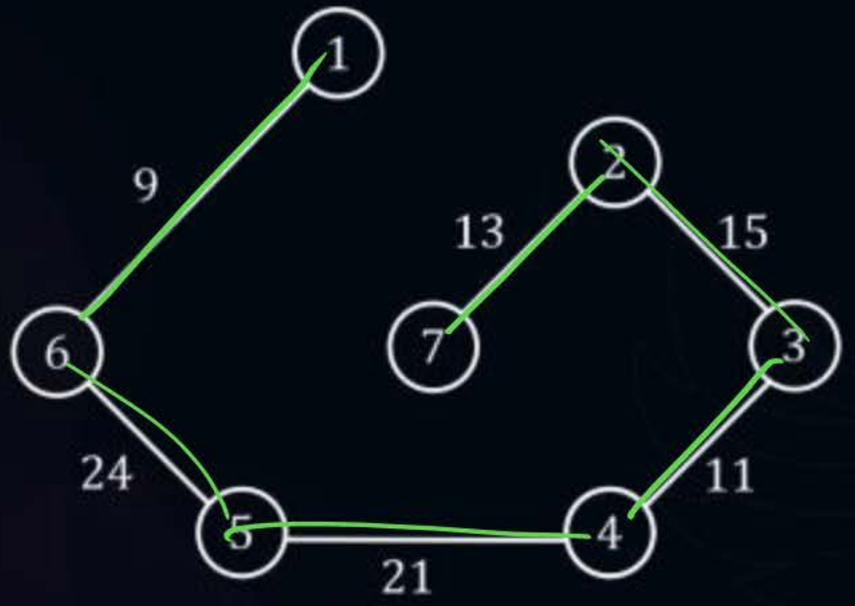






Topic: Sets

Min cost: = 0 + 9 + 11 + 13 + 15 + 21 + 24 = 93





Topic: Sets

1. (u, v) = (1, 6)

u v

j = find (1) = 1

k = find (6) = 6

j ≠ k

t [1, 1] =1

t [1, 2] =6

Union (1, 6)

2. (u, v) = (3, 4)

j = find (3) = 3

k = find (4) = 4

j ≠ k

t [2, 1] =3

t [2, 2] =4

Union (3, 4)



Topic: Sets

3. (u, v) = (2, 7)

j = find (2) = 2

k = find (7) = 7

j ≠ k

t [3, 1] =2

t 3, 2] =7

Union (2, 7)



4. (u, v) = (2, 3)

j = find (2) = 7

k = find (3) = 4

j ≠ k

t [4, 1] = 2

t [4, 2] = 3

Union (7, 4)





Topic: Sets

5. $(u, v) = (7, 4)$

$j = \text{find}(7) = 4$

$k = \text{find}(4) = 4$

$j = k$

(cycle)

6. $(u, v) = (5, 4)$

$j = \text{find}(5) = 5$

$k = \text{find}(4) = 4$

$j \neq k$

$t[5, 1] = 5$

$t[5, 2] = 4$

Union (5, 4)





Topic: Sets

7. (u, v) = (7, 5)

j = find (7) = 4
k = find (5) = 4

j = k
(cycle)

8. (u, v) = (5, 6)

j = find (6) = 6
k = find (5) = 4

j ≠ k
t [6, 1] = 6
t [6, 2] = 5
Union (6, 4)



Topic: Sets

Time and space complexity analysis of Kruskal (overall).

e = number of edges in given graph

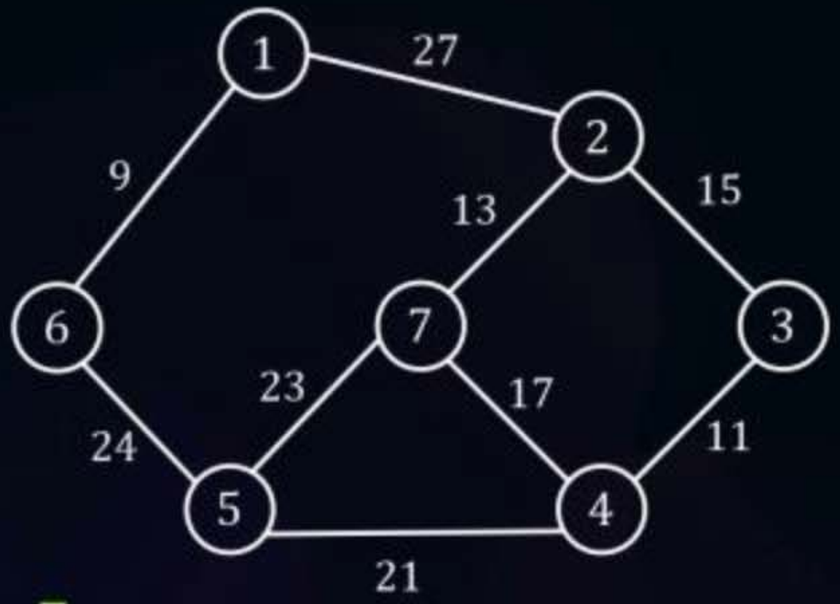
1. Time complexity = $O(e^* \log e)$
2. Space complexity = $O(e + v)$
   = $O(e + n)$



Topic: Sets

Kruskal logic:

Given a graph:



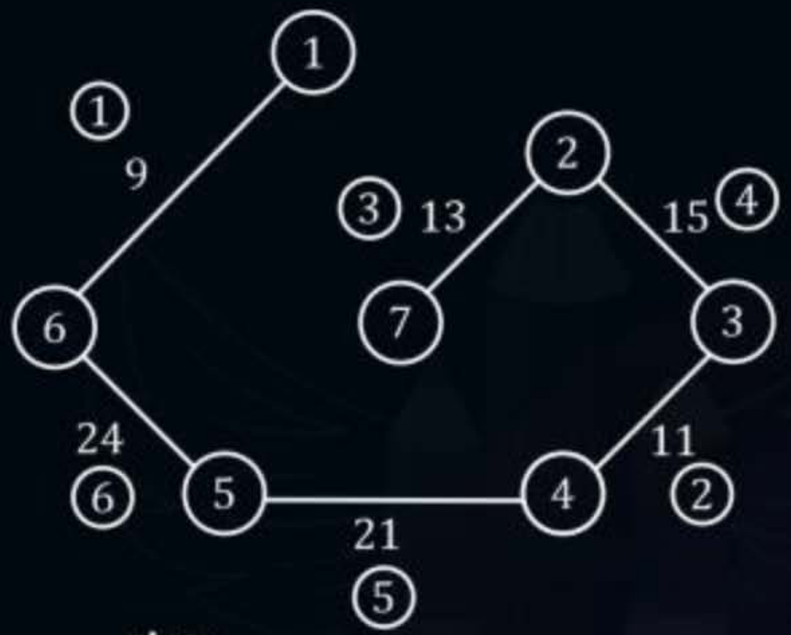
n = |v| = 7

MCST edges = n –1

= 6

Cost of MCST

= 9 + 11 + 13 + 15 +21 + 24 = 93



stop

[MCQ]

HW

#Q. Consider the following array:

23	32	45	69	72	73	89	97

Which algorithm out of the following options uses the least number of comparisons (among the array elements) to sort the above array in ascending order?

A. Quick sort using the last elements as pivot

B. Selection sort

C. Merge sort

D. Insertion sort

[MCQ]

#Q. The median of n elements can be founded can be found in 0 (n) time. Which one of the following is correct about the complexity of quick sort, in which remains is selected as pivot?

A $\Theta(n)$

B $\Theta(n \log n)$

C $\Theta(n^2)$

D $\Theta(n^3)$



2 mins Summary

→ Set

→ Kruskal



THANK - YOU